

Generate an equivalent algebraic expression where each variable only appears once.

1. $(5h^{-6}m^3)(6h^5m^7)^2$

work

**Equivalent
Algebraic
Expression**

2. $(-7xy^{-4})(9x^5z^3)(3x^4y^2z^{-1})$

work

**Equivalent
Algebraic
Expression**

3. $(-8h^3k^{-5})^2 \cdot (10h^0k^6)$

work

**Equivalent
Algebraic
Expression**

4. $\frac{(2c^3)^5(4c^4)}{(8c^6)^2}$

work

**Equivalent
Algebraic
Expression**

5. Use the laws of exponents to complete the equivalent expression for 5.6^{2x} .

$$\left(\boxed{} \right)^x$$

6. Use the laws of exponents to complete the equivalent expression for 3^{5x-3} .

$$\frac{\left(\boxed{} \right)^x}{\boxed{}}$$

7. Find the sum.

$$(x + 3)(2x^5) + 7x^5$$

work

Sum

8. Find the product.

$$(k + 8)(k - 11)$$

work

Product

9. Find the sum.

$$-2x(x + 9) + (x + 8)(7 - 3x)$$

work

Sum

10. Find the difference.

$$\frac{2}{3}(7y^3 - 9y) - \left(\frac{11}{3}y^3 - \frac{2}{3}y^2 + \frac{17}{3}\right)$$

work

Difference

11. Find the product.
 $(2x + 9)(5x - 8)$

work

Product	
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12. Find the product.
 $(x + 7)^2$

work

Product	
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13. Find the product.

$$\frac{4}{5}b^6(11b - 15)$$

work

Product

14. Find the quotient where $h \neq 0$.

$$(24h^9 + 32h^{15}) \div (-8h^3)$$

work

Quotient

15. Find the product.

$$(x - 4)(x - 7)$$

work

Product

16. Find the quotient where $w \neq 0$ and $y \neq 0$.

$$\frac{16w^5y^{11} + 28w^2y^5 - 36w^3y^3}{4wy^3}$$

work

Quotient